# Protobuf protocol example

```
syntax = "proto3";

message Control {
  enum InputMux {
    INPUT_MUX_UNSPECIFIED = 0;
    ADC_CHANNEL_0 = 1;
    ADC_CHANNEL_1 = 2;
    ADC_CHANNEL_6 = 7;
    ADC_CHANNEL_7 = 8;
  };

  enum DspOutputMux {
    DSP_MUX_UNSPECIFIED = 0;
    DIGITAL_DOWN_CONVERTER = 1;
    // Other values
  }

  enum SyncPulseMux {
    SYNC_MUX_UNSPECIFIED = 0;
    LOW_LOGIC_LEVEL = 1;
    HIGH_LOGIC_LEVEL = 2;
  }

  enum TxOutputMux {
    TX_MUX_UNSPECIFIED = 0;
    DIGITAL_UP_CONVERTER = 1;
  }

  bool dsp_enabled = 1;
  bool pilot_enabled = 2;
  bool probe_enabled = 3;

  uint32 overview_mode = 4;
  uint32 tx2rx_latency = 5;
  uint32 dec_factor_code = 6;

  DspOutputMux dsp_output_mux = 7;
  SyncPulseMux sync_pulse_mux = 8;
  TxOutputMux tx_output_mux = 9;

  repeated InputMux input_mux_rx = 10;
  /* Array, max size 64 MB (Safe limit, 2 GB without limits)
   * But atomic, cannot give only one field
   * For Non-atomic fields use MAP
   */
}

message Tasks {
```

```
    message QueueItem {
      bool active = 1;
      bool mode_switch = 2;
      int32 index = 3;
      int32 tag = 4;
      int32 sm = 5;
      int32 az_code = 6;
      int32 el_code = 7;
      int32 fr_code = 8;
    }

    bool sm_ctrl = 1;
    bool steering_ctrl = 2;
    repeated QueueItem que = 3;
  }

  message OverviewModes {
    message PacketOfPulses {
      bool steering_ctrl = 1;
      bool prb_disable = 2;
      bool active = 3;
      bool orthogonal = 4;
      bool qcw = 5;
      uint32 bid = 6;
      uint32 index = 7;
      uint32 nor = 8;
      double prp_us = 9;
      double phase_deg = 10;
      uint32 tag = 11;
    }

    message Modes {
      uint32 mid = 1;
      string description = 2;
      repeated PacketOfPulses packet_of_pulses = 3;
    }

    repeated Modes overview_modes = 1;
  }

  message PulseBursts {
    message PilotBurst {
      bool active = 1;
      bool sap = 2;
      double amplitude_dB = 3;
      double doppler_kHz = 4;
      uint32 index = 5;
      double pause_us = 6;
      uint32 sid = 7;
      double width_us = 8;
    }

    message ProbeBurst {
      bool active = 1;
```

```
    bool window = 2;
    double fpf_ns = 3;
    double rpf_ns = 4;
    double if_MHz = 5;
    uint32 index = 6;
    double pause_us = 7;
    uint32 sid = 8;
    double width_us = 9;
  }

  message InnerObj {
    uint32 bid = 1;
    string description = 2;
    repeated PilotBurst pilot_burst = 3;
    repeated ProbeBurst probe_burst = 4;
  }

  repeated InnerObj burst = 1;
}

message SignalIds {
  message Signal {
    double scale = 1;
    repeated double amplitude_coefs = 2;
    repeated double phase_coefs = 3;
    string description = 4;
  }

  message XCorr {
    double scale = 1;
    repeated double amplitude_coefs = 2;
    repeated double phase_coefs = 3;
  }

  message GaussWindow {
    double mean = 1;
    double sigma = 2;
    double hold = 3;
    double stop = 4;
  }

  message Weight {
    double scale = 1;
    repeated double filter_coefs = 2;
  }

  message Ivm {
    double pause_us = 1;
    double width_us = 2;
    double start_pause_us = 3;
    double start_width_us = 4;
  }

  message InnerObj {
```

```
    uint32 sid = 1;
    Signal signal = 2;
    XCorr x_corr = 3;
    GaussWindow gauss_window = 4;
    Weight weight = 5;
    Ivm ivm = 6;
  }

  repeated InnerObj signal_ids = 1;
}

message MainConfig {
  Control ctrl = 1;
  Tasks tasks = 2;
  OverviewModes overview_modes = 3;
  PulseBursts pulse_bursts = 4;
  SignalIds signal_ids = 5;
}

message BasicResp {
  uint32 id = 1;
  string message = 2;
}
```